Supporting Information for: "A Universal Quantum Descriptor for Electron Correlation in Molecular Systems: The F_{bond} Framework"

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November 2025 — Version 2.0

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1 Methodological Consistency (Version 2.0 Update)

1.1 Evolution from Version 1.0 to 2.0

The original manuscript (v1.0) employed two different approaches for extracting entanglement entropy:

1. **VQE-based approach** (NH_3, H_2O): Single-qubit entanglement entropy from variational quantum eigensolver statevector
2. **FCI-based approach** (CH_4, C_2H_4, C_2H_2): Natural orbital entanglement from frozen-core FCI wavefunction

These two approaches measure fundamentally different quantities:

- **Qubit entropy**: Entanglement in computational basis (Jordan-Wigner or parity mapping)
- **Natural orbital entropy**: Entanglement in eigenbasis of one-particle density matrix

Version 2.0 employs consistent frozen-core FCI natural orbital analysis across all molecules, yielding comparable entropy measures and cleaner regime separation.

1.2 Impact on Results

Table 1: Comparison of F_{bond} $valuesbetweenversions.$

Molecule	v1.0 F_{bond}	v2.0 F_{bond}	Change	v2.0 Classification
NH_3	5.2×10^{-4}	0.0321	60×	Weak correlation
H_2O	9.6×10^{-4}	0.0352	37×	Weak correlation
CH_4	0.0400	0.0400	—	Weak correlation
H_2	0.0425	0.0425	—	Weak correlation
C_2H_2	0.0651	0.0651	—	Strong correlation
N_2	0.0667	0.0667	—	Strong correlation
C_2H_4	0.0718	0.0718	—	Strong correlation

Key finding: With consistent methodology, all σ-only molecules cluster at $F_{\text{bond}} \approx 0.035$, revealing that bond type (σ vs π) is the dominant factor rather than bond polarity.

2 Complete Computational Details

2.1 Software and Hardware

All calculations were performed using:

- **Software:** PySCF v2.X [1, 2]
- **Hardware:** AMD Ryzen 9 9950X (16 cores @ 3.5 GHz), 64 GB RAM
- **Operating System:** Linux (6.16.8-1-MANJARO-Xfce)

2.2 Basis Set Details

STO-3G: Minimal basis set

- H: 1 basis function (1s)
- C: 5 basis functions (1s, 2s, $2p_x$, $2p_y$, $2p_z$)
- N: 5 basis functions (1s, 2s, $2p_x$, $2p_y$, $2p_z$)
- O: 5 basis functions (1s, 2s, $2p_x$, $2p_y$, $2p_z$)

6-31G: Split-valence basis set (used for H_2O validation)

- H: 2 basis functions
- O: 13 basis functions

2.3 Frozen-Core Treatment

For molecules containing carbon, nitrogen, or oxygen, core orbitals were frozen at the Hartree-Fock level:

Table 2: Frozen-core configuration for each molecule.

Molecule	Core Orbitals	Core Electrons	Active Space
H_2	None	0	(2e, 2o)
NH_3	N 1s	2	(8e, 7o)
H_2O	O 1s	2	(8e, 6o)
CH_4	C 1s	2	(8e, 8o)
C_2H_4	2× C 1s	4	(12e, 13o)
C_2H_2	2× C 1s	4	(10e, 11o)
N_2	2× N 1s	4	(10e, 10o)

Rationale: Core electrons (1s orbitals) do not participate in chemical bonding and their correlation with valence electrons can create artificial fractional occupations in minimal basis sets. Frozen-core treatment ensures F_{bond} $measuresonlychemicalcorrelationrelevanttobonding.$

2.4 Convergence Criteria

- Hartree-Fock: Energy convergence 10^{-10} Ha, density matrix 10^{-8}
- CCSD: Energy convergence 10^{-9} Ha, amplitude 10^{-7}
- FCI (CASCI): Energy convergence 10^{-10} Ha, wavefunction 10^{-8}

3 Molecular Geometries

All geometries were optimized at the B3LYP/6-311G** level or taken from experimental structures.

3.1 Hydrogen (H_2)

```
H  0.0000  0.0000  0.0000
H  0.0000  0.0000  0.7414
```

Bond length: 0.7414 Å (experimental equilibrium)

3.2 Nitrogen (N_2)

```
N  0.0000  0.0000  0.0000
N  0.0000  0.0000  1.0980
```

Bond length: 1.098 Å (experimental equilibrium)

3.3 Ammonia (NH_3)

```
N  0.0000  0.0000  0.0000
H  0.0000  0.9400  0.3800
H  0.8137 -0.4700  0.3800
H -0.8137 -0.4700  0.3800
```

N-H bond length: ∼1.01 Å, H-N-H angle: ∼107° (pyramidal, C_{3v} symmetry)

3.4 Water (H_2O)

```
O  0.0000  0.0000  0.0000
H  0.0000  0.7570  0.5870
H  0.0000 -0.7570  0.5870
```

O-H bond length: 0.96 Å, H-O-H angle: 104.5° (bent, C_{2v} symmetry)

3.5 Methane (CH_4)

```
C  0.0000  0.0000  0.0000
H  0.6276  0.6276  0.6276
H -0.6276 -0.6276  0.6276
H  0.6276 -0.6276 -0.6276
H -0.6276  0.6276 -0.6276
```

C-H bond length: 1.087 Å (tetrahedral, T_d symmetry)

3.6 Ethylene (C_2H_4)

```
C  0.0000  0.0000  0.6695
C  0.0000  0.0000 -0.6695
H  0.0000  0.9289  1.2321
H  0.0000 -0.9289  1.2321
H  0.0000  0.9289 -1.2321
H  0.0000 -0.9289 -1.2321
```

C=C bond length: 1.339 Å, C-H bond length: 1.086 Å (planar, D_{2h} symmetry)

3.7 Acetylene (C_2H_2)

```
C  0.0000  0.0000  0.6010
C  0.0000  0.0000 -0.6010
H  0.0000  0.0000  1.6630
H  0.0000  0.0000 -1.6630
```

C≡C bond length: 1.202 Å, C-H bond length: 1.062 Å (linear, $D_{\infty h}$ symmetry)

4 Complete Natural Orbital Analysis

4.1 Hydrogen (H_2)

Table 3: Natural orbital occupations for H_2 (STO-3G, full FCI).

Orbital	Occupation	Deviation	Entropy (nats)
NO 1 (σ)	1.9760	0.0240	0.0681
NO 2 (σ^*)	0.0240	0.0240	0.0667

Classification: Weak correlation. H_2 shows 2.4% fractional character typical of σ-only systems.

4.2 Ammonia (NH_3)

Table 4: Natural orbital occupations for NH_3 (STO-3G, frozen-core FCI).

Orbital	Occupation	Deviation	Entropy (nats)
NO 1–4 (σ)	1.968–1.971	0.029–0.032	0.079–0.084
NO 5 (highest)	1.9695	0.0305	0.0649
NO 6–7 (σ^*)	0.030–0.031	0.030–0.031	0.080–0.083

Classification: Weak correlation. Despite high bond polarity ($\Delta\chi = 0.9$), NH_3 shows only 3% fractional character, clustering with other σ-only molecules. $S_{E,\max} = 0.0649 nats$.

4.3 Water (H_2O)

Table 5: Natural orbital occupations for H_2O (STO-3G, frozen-core FCI).

Orbital	Occupation	Deviation	Entropy (nats)
NO 1–4 (σ)	1.968–1.972	0.028–0.032	0.076–0.085
NO 5 (highest)	1.9702	0.0298	0.0706
NO 6 (σ^*)	0.0297	0.0297	0.0793

Classification: Weak correlation. Despite highest bond polarity in dataset ($\Delta\chi = 1.4$), H_2O shows only 3% fractional character. $S_{E,\max} = 0.0706 nats, nearly identical to NH_3 and CH_4$.

Table 6: Natural orbital occupations for CH_4 (STO-3G, frozen-core FCI).

Orbital	Occupation	Deviation	Entropy (nats)
NO 1	1.9852	0.0148	0.0436
NO 2 (C-H)	1.9761	0.0239	0.0647
NO 3 (C-H)	1.9761	0.0239	0.0647
NO 4 (C-H)	1.9761	0.0239	0.0647
NO 5 (σ^*)	0.0222	0.0222	0.0611
NO 6 (σ^*)	0.0222	0.0222	0.0611
NO 7 (σ^*)	0.0222	0.0222	0.0611
NO 8 (σ^*)	0.0197	0.0197	0.0554

4.4 Methane (CH_4)

Classification: Weak correlation. CH_4 shows 2.4% fractional character, clustering with NH_3, H_2O, and H_2 despite varying bond polarities.

4.5 Ethylene (C_2H_4)

Table 7: Natural orbital occupations for C_2H_4 (STO-3G, frozen-core FCI).

Orbital	Occupation	Deviation	Entropy (nats)
NO 1–5 (σ)	1.974–1.984	0.016–0.026	0.046–0.069
NO 6 (π)	1.8823	0.1177	0.2238
NO 7 (π^*)	0.1181	0.1181	0.2243
NO 8–13 (σ^*)	0.010–0.026	0.010–0.026	0.030–0.068

Note: π/π^* orbitals show strong fractional occupations (11.8% depletion/population), characteristic of π–π^* correlation.

Classification: Strong correlation. The π system shows 4× more fractional character than σ-only molecules.

4.6 Acetylene (C_2H_2)

Table 8: Natural orbital occupations for C_2H_2 (STO-3G, frozen-core FCI).

Orbital	Occupation	Deviation	Entropy (nats)
NO 1–3 (σ)	1.977–1.981	0.019–0.023	0.053–0.061
NO 4 (π_x)	1.9158	0.0842	0.1745
NO 5 (π_y)	1.9158	0.0842	0.1745
NO 6 (π_x^*)	0.0843	0.0843	0.1747
NO 7 (π_y^*)	0.0843	0.0843	0.1747
NO 8–10 (σ^*)	0.010–0.031	0.010–0.031	0.029–0.081

Note: Two degenerate π systems show identical fractional occupations (8.4%).

Classification: Strong correlation. Triple bond shows similar π correlation to double bond.

Table 9: Typical runtimes on AMD Ryzen 9 9950X (16 cores).

Molecule	Basis	HF (s)	CCSD (s)	FCI (s)
H_2	STO-3G	<1	1	1
NH_3	STO-3G	<1	8	90
H_2O	STO-3G	<1	5	60
CH_4	STO-3G	<1	5	10
N_2	STO-3G	1	10	15
C_2H_4	STO-3G	1	45	900
C_2H_2	STO-3G	1	8	120

5 Computational Timings

Note: C_2H_4 and NH_3 require longer times due to larger active spaces.

6 Validation: CCSD vs. FCI Energy Comparison

Table 10: Energy differences between frozen-core CCSD and FCI.

Molecule	$\lvert E_{\text{FCI}} - E_{\text{CCSD}} \rvert$ (mHa)	Agreement
H_2	0.3	Excellent
NH_3	1.2	Excellent
H_2O	0.8	Excellent
CH_4	0.5	Excellent
N_2	3.8	Good
C_2H_4	2.1	Good
C_2H_2	1.5	Excellent

Note: All differences < 5 mHa, confirming CCSD captures majority of correlation energy.

7 Regime Classification Summary

Table 11: Complete F_{bond} $classificationwithconsistentfrozen - coreFCImethodology.$

Molecule	Bond Type	F_{bond}	Fractional %	Regime
	Weak Correlation (σ-only systems)			
NH_3	N-H σ (3×)	0.0321	3.0%	DFT/MP2
H_2O	O-H σ (2×)	0.0352	3.0%	DFT/MP2
CH_4	C-H σ (4×)	0.0400	2.4%	DFT/MP2
H_2	H-H σ	0.0425	2.4%	DFT/MP2
	Strong Correlation (π-containing systems)			
C_2H_2	C≡C (σ+2π)	0.0651	8.4%	CCSD+
N_2	N≡N (σ+2π)	0.0667	7.7%	CCSD+
C_2H_4	C=C (σ+π)	0.0718	11.8%	CCSD+

Key Insight: All σ-only molecules cluster at $F_{\text{bond}} \approx 0.035$ with 2–3% fractional character, regardless of bond polarity ($\Delta\chi = 0$ to 1.4). All π-containing molecules show $F_{\text{bond}} \approx 0.070$ with

8–12% fractional character.

8 Code Availability

Complete Jupyter notebooks with detailed implementations are provided as supplementary files:

- `NH3_Frozen_Core_FCI.ipynb`
- `H2O_Frozen_Core_FCI.ipynb`
- `CH4_Frozen_Core_STO3G.ipynb`
- `C2H4_Frozen_Core_STO3G.ipynb`
- `C2H2_Frozen_Core_STO3G.ipynb`

All notebooks use PySCF v2.X and produce CSV output files with complete results.

References

[1] Sun, Q.; et al. PySCF: the Python-based simulations of chemistry framework. *WIREs Comput. Mol. Sci.* **2018**, *8*, e1341.

[2] Sun, Q.; et al. Recent developments in the PySCF program package. *J. Chem. Phys.* **2020**, *153*, 024109.